The Maximum Number of Sides in the Intersection of Two Unit Regular Polygons

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Abstract

For integers $n, m \geq 3$, let $K(n, m)$ be the largest possible number of sides of the intersection of a unit-side-length regular n-gon and a unit-side-length regular m-gon, under arbitrary rigid motions and with nonempty interior. This is the maximum-edge subproblem in Problem 93 of Kagey's open problem collection. We determine $K(n, m)$ exactly. Besides $K(n, n) = 2n$, the low rows are

$$K(3, m) = 5 \ (m > 3), \qquad K(4, m) = 7 \ (m > 4), \qquad K(5, m) = 8 \ (m > 5).$$

For $6 \leq n < m$, put $d = m - n$ and

$$c_q = \left\lfloor \frac{q(q-2)}{2} \right\rfloor \qquad (q \geq 4).$$

Then

$$K(n, m) = n + \max\Big(\{3\} \cup \{q \geq 4 : n \geq dc_q + \kappa_d(q)\}\Big),$$

where

$$\kappa_d(q) = \begin{cases} 1, & d \geq 4, \\ 1, & d = 3,\ q = 4, \\ 2, & d = 3,\ q \geq 5, \\ 2, & d = 2, \\ 2, & d = 1,\ q = 4, \\ 3, & d = 1,\ q \in \{5, 6, 7, 8, 10\}, \\ 4, & d = 1,\ q = 9 \text{ or } q \geq 11. \end{cases}$$

The proof has three parts. First, for $n \geq 6$ a global crossing-index argument reduces every configuration with excess at least four to a positive branch and eliminates the possible gain-two endpoint defects without suppressing gap parameters. Second, a sparse-to-symmetric domination theorem shows that arbitrarily long gain-one excursions cannot improve on the centered alternating threshold; the converse is proved by a global realization argument, not by local interpolation alone. Third, the exact trigonometric threshold is integerized by a Peano-kernel argument. Only finitely many boundary signs remain; one exact rational-interval Python certificate supplies all required signs and retains several redundant boundary cross-checks.

1 Introduction

Let P_k be a regular k-gon of side length 1. Peter Kagey's Problem 93 asks for the possible combinatorial classes of intersections of two unit regular polygons and, as a related question, asks

for the largest number of sides that such an intersection can have [1]. The present paper solves this maximum-edge problem.

The intersection of two convex polygons is again convex and has at most the sum of the two numbers of sides; this elementary linear-complexity fact is standard in computational geometry [2, Sec. 7.6]. What is nontrivial here is the rigidity imposed by equal side lengths and regularity. Those constraints force a sharp staircase law rather than the trivial upper bound $n + m$.

We write

$$K(n, m) := \max \#E(P_n \cap gP_m),$$

where g ranges over rigid motions for which the intersection has nonempty interior. Clearly

$$K(n, m) = K(m, n).$$

The main theorem is stated in Section 9. The proof is organized as follows.

(i) Sections 2–3 establish source counting, the low-row upper bounds, and the diagonal case.

(ii) Sections 4–5 prove, for $n \geq 6$, the global reduction from arbitrary configurations to exact even/odd symmetric trigonometric thresholds. Endpoint gaps are retained explicitly, and long gain-one excursions are never identified with single teeth.

(iii) Section 6 proves global realizability of those thresholds and supplies the universal $+3$ construction, completing the rows $n = 4, 5$ as well.

(iv) Sections 7–8 convert the thresholds into the integer staircase using a Peano kernel and one finite exact certificate.

Support functions and elementary facts about convex polygons are used throughout; see Schneider [3] for standard background. All nonstandard ingredients required for the proof are established below.

2 Basic notation and source counting

For $k \geq 3$ set

$$R_k = \frac{1}{2\sin(\pi/k)}, \qquad \rho_k = \frac{1}{2}\cot\frac{\pi}{k}.$$

Thus R_k and ρ_k are the circumradius and inradius of the unit regular k-gon. We also write

$$u(\theta) = (\cos\theta, \sin\theta).$$

Lemma 2.1 (general position reduction)**.** *The maximum $K(n, m)$ may be computed using configurations in which no vertex lies on the other polygonal boundary, no two supporting lines from the two polygons coincide, and all boundary intersections are transversal.*

Proof. Fix a configuration with nonempty interior. Choose one relative-interior point on each positive-length edge of the intersection. Each such point satisfies finitely many strict half-plane inequalities except possibly when the same supporting line is supplied by both polygons. A sufficiently small generic translation and rotation preserves all strict inequalities; in the coincident-support case, one of the two nearby supports still contributes a positive-length edge. Thus a generic perturbation produces at least as many positive-length intersection edges. Since the edge count is integer-valued and bounded by $n + m$, it suffices to prove upper bounds in general position and to construct lower-bound examples there. □

In general position let A, B be two convex polygons and $Q = A \cap B$. Let a be the number of vertices of A in $\operatorname{int} B$, let b be the number of vertices of B in $\operatorname{int} A$, and let c be the number of proper crossings of ∂A and ∂B.

Lemma 2.2 (vertex–crossing identity)**.**

$$\#E(Q) = a + b + c.$$

Proof. Every vertex of Q is uniquely of one of the following three types: a vertex of A lying in $\operatorname{int} B$, a vertex of B lying in $\operatorname{int} A$, or a transversal crossing of one edge of A and one edge of B. Conversely every such point is a vertex of Q. A convex polygon has equal numbers of vertices and edges. □

3 Low-row upper bounds and the diagonal

We first isolate the rows that will not use the global compression argument.

3.1 A side-gap metric

Let E_i, E_j be edges of a unit regular m-gon and put

$$q(i,j) = \min\{|i-j|, m-|i-j|\}.$$

Lemma 3.1 (side-gap metric)**.** *For $2 \le q \le \lfloor m/2 \rfloor$,*

$$\operatorname{dist}(E_i, E_j) = \frac{\sin((q-1)\pi/m)}{\sin(\pi/m)}.$$

In particular, two nonadjacent edges are at distance at least 1*.*

Proof. By rotation take $E_i = E_0$ and $E_j = E_q$. Put $\theta = \pi/m$ and

$$V_k = R_m(\cos 2k\theta, \sin 2k\theta).$$

In the direction $w = (V_q - V_1)/|V_q - V_1|$, the projections of both edge directions are $\cos(q\theta) \ge 0$. Hence the maximal projection on E_0 occurs at V_1 and the minimal projection on E_q at V_q. Thus the distance equals $|V_q - V_1|$, which is the displayed chord length. The final assertion follows from $(q-1)\pi/m \in [\pi/m, \pi/2]$. □

Lemma 3.2 (diameter-side clique principle)**.** *Let C be a compact convex set of diameter D. If two edges of a unit regular m-gon contribute positive-length edges to $C \cap P_m$ in general position, then their mutual distance is strictly less than D.*

Proof. Choose relative-interior points x, y on the two contributed segments. They lie in $\operatorname{int} C$. Two interior points cannot realize the diameter of a convex body with nonempty interior, because the segment can be extended slightly at both ends while remaining in C. Hence $|x - y| < D$. □

For fixed $s \ge 2$, the quantity

$$d_s^{(N)} := \frac{\sin(s\pi/N)}{\sin(\pi/N)}$$

is strictly increasing in N whenever $s \le N/2$. Indeed, differentiating its logarithm after putting $x = \pi/N$ reduces this to the strict decrease of $x \cot x$ on $(0, \pi)$.

3.2 The triangle row

Theorem 3.3.

$$K(3,m) = \begin{cases} 6, & m = 3, \\ 5, & m \geq 4. \end{cases}$$

Proof. A unit equilateral triangle has diameter 1. Lemmas 3.1 and 3.2 imply that for $m \geq 4$ the contributing P_m-edges are pairwise adjacent, hence there are at most two; therefore $K(3,m) \leq 5$.

For the lower bound, take the unit triangle with vertices

$$(-1/2, 0), \quad (1/2, 0), \quad (0, \sqrt{3}/2)$$

and put a vertex of P_m at $O = (0, \sqrt{3}/2 - 1/4)$, with its interior-angle bisector pointing vertically downward. The half interior angle of P_m is at least $\pi/4$, so the two bottom triangle vertices lie strictly inside that corner and the top vertex lies outside. The whole triangle lies in the open unit disk centered at O, whereas every nonincident edge of P_m is at distance at least 1 from O. Hence the vertex counts are $(a,b,c) = (2,1,2)$ and Lemma 2.2 gives 5. For $m = 3$, two concentric oppositely oriented equilateral triangles have a hexagonal intersection. □

3.3 Square and pentagon upper bounds

For $m \geq 6$, if two contributing P_m-edges have cyclic gap at least 3, then their distance is at least

$$2\cos(\pi/m) \geq \sqrt{3}.$$

The unit square and unit regular pentagon have diameters $\sqrt{2}$ and $(1+\sqrt{5})/2$, both strictly below $\sqrt{3}$. Thus, in either row, the contributing P_m-edge indices have pairwise cyclic distance at most 2. A cyclic set of length $m \geq 6$ with this property has at most three elements: if four existed and the four cyclic gaps were $d_1, \ldots, d_4 \geq 1$, one opposite pair would force two consecutive gaps to be 1, and the other opposite pair a third gap to be 1; the remaining gap is $m - 3 \geq 3$, a contradiction. Therefore

$$K(4,m) \leq 7, \qquad K(5,m) \leq 8 \qquad (m \geq 6). \tag{3.1}$$

The remaining square–pentagon upper bound is also analytic.

Lemma 3.4. $K(4,5) \leq 7$.

Proof. Let $v_0, \ldots, v_4$ be the outward unit normals of the pentagon and put $\tau = (1+\sqrt{5})/2$. Then

$$v_0 + v_1 + \tau v_3 = 0.$$

If the three corresponding pentagon support lines all met a unit square Q, then with ρ_5 the pentagon inradius,

$$(2+\tau)\rho_5 \leq h_Q(v_0) + h_Q(v_1) + \tau h_Q(v_3).$$

Put $w_0 = v_0, w_1 = v_1, w_3 = \tau v_3$. Since $w_0 + w_1 + w_3 = 0$, for either coordinate direction e of the square,

$$\frac{1}{2}\sum_i |w_i \cdot e| = \max_i |w_i \cdot e|.$$

Optimizing over the square orientation therefore bounds the right side by

$$M = \sqrt{1 + \tau^2 + 2\tau\sin(\pi/5)}.$$

Since $\sin^2(\pi/5) = (5-\sqrt{5})/8 < 9/25$, hence $\sin(\pi/5) < 3/5$,

$$M^2 < 2 + \frac{11}{5}\tau,$$

whereas the exact identity $\rho_5^2 = (5+2\sqrt{5})/20$ gives

$$((2+\tau)\rho_5)^2 = \frac{7+11\tau}{4} > 2 + \frac{11}{5}\tau.$$

Contradiction. Any four pentagon edges contain a rotated copy of the triple $\{0,1,3\}$, so at most three pentagon edges contribute. □

Thus $K(4,m) \le 7$ for every $m \ge 5$, and $K(5,m) \le 8$ for every $m \ge 6$. Their matching lower bounds will follow from the universal $+3$ constructions in Section 6.

3.4 The diagonal

Theorem 3.5. *For every $n \ge 3$,*

$$K(n,n) = 2n.$$

Proof. The source-edge bound gives $K(n,n) \le 2n$. Take two concentric unit regular n-gons and rotate one by a sufficiently small nonzero angle. Their $2n$ support normals strictly interlace and all supports have the same distance ρ_n from the common center, so none of the $2n$ supporting half-planes is redundant. The intersection has $2n$ sides. □

4 Excursions and the global index-gap theorem

From now until Section 6 assume

$$6 \le n < m.$$

Set

$$A = P_n, \qquad B = z + R_\varphi P_m, \qquad \alpha = \frac{2\pi}{n}, \quad \beta = \frac{2\pi}{m}, \quad \delta = \alpha - \beta > 0.$$

Lemma 4.1 (radius separation)**.** *For $5 \le n < m$,*

$$\rho_m > R_n.$$

Proof. Since ρ_k increases with k, it suffices to prove $\rho_{n+1} > R_n$. For $n = 5$ this is equivalent to $7 > 3\sqrt{5}$. For $n \ge 6$, put $x = \pi/n$, $y = \pi/(n+1)$, $u = x - y$. Then

$$\sin x \cos y - \sin y = \sin u - 2\sin^2(x/2)\sin y.$$

Now $\sin u > u - u^3/6 > 5u/6$, while

$$2\sin^2(x/2)\sin y < \frac{x^2 y}{2} = u\frac{\pi^2}{2n} < \frac{5}{6}u.$$

Hence the difference is positive. □

If a B-edge with outward normal v contributes to the intersection boundary, then its support line meets A, so

$$\rho_m + z \cdot v \le h_A(v) \le R_n < \rho_m.$$

Thus $z \cdot v < 0$ and all contributing B-edge normals lie in one open semicircle. Cut the cyclic indices through its complementary arc and lift all relevant indices to integers. The A-indices are lifted simultaneously by the common boundary order. In particular, if two crossings lie on the same physical A-edge but on opposite sides of the chosen cyclic cut, their lifted A-indices differ by n rather than by 0; all index differences below are understood in this lifted sense.

A component of $\partial B \cap \operatorname{int} A$ is called a *B-inside excursion*. For excursion E_i, let b_i be its number of interior B-vertices and let s_i be the number of interior A-vertices on the replaced lifted A-arc between its two crossings (equivalently, the lifted A-edge index increment). Define

$$g_i = b_i - s_i + 2. \tag{4.1}$$

As the excursion replaces $s_i - 1$ complete A-edges by $b_i + 1$ B-edge pieces,

$$\#E(A \cap B) - n = \sum_i g_i. \tag{4.2}$$

If a crossing P lies on A_i and B_j, write $k(P) = j - i$. The following is the global discrete separation estimate.

Theorem 4.2 (global index-gap theorem)**.** *Let*

$$P \in A_i \cap B_j, \qquad Q \in A_{i+p} \cap B_{j+r}$$

be crossings in their common lifted order. Then

$$r \le p + 1, \qquad k(Q) - k(P) \le 1. \tag{4.3}$$

Proof. The common translation makes the corresponding chord vectors in A and in the untranslated P_m equal, hence their lengths agree. Suppose $r \ge p + 2$. If $p = 0$, the two A-points lie strictly inside one unit edge and are less than 1 apart, while the two B-edges are nonadjacent and have distance at least 1 by Lemma 3.1. If $1 \le p$ and $p + 1 \le n/2$, the A-distance is at most $d^{(n)}_{p+1}$, whereas the B-distance is at least

$$d^{(m)}_{r-1} \ge d^{(m)}_{p+1} > d^{(n)}_{p+1}.$$

If $p + 1 > n/2$, let $s = \lfloor n/2 \rfloor$. The A-distance is at most $d^{(n)}_s$. The active B-window is strictly shorter than a semicircle, so $r - 1 < m/2$; since $r - 1 > s$,

$$d^{(m)}_{r-1} > d^{(m)}_s > d^{(n)}_s.$$

Every case is contradictory. □

Let P_i^-, P_i^+ be the start and end crossings of E_i. Let the lifted difference "A-normal angle minus B-normal angle" be $-\ell_i < 0$ at the start and $r_i > 0$ at the end. Between E_i and E_{i+1} let the A- and B-edge index increments be a_i, c_i. Normal order gives

$$r_i + \ell_{i+1} = c_i\beta - a_i\alpha > 0,$$

so $c_i \ge a_i + 1$. Theorem 4.2 applied to the two gap crossings gives the reverse inequality, hence

$$\boxed{c_i = a_i + 1, \qquad r_i + \ell_{i+1} = \beta - a_i\delta \le \beta.} \tag{4.4}$$

Across an excursion,

$$k(P_i^+) - k(P_i^-) = b_i - s_i = g_i - 2$$

and

$$\boxed{\ell_i + r_i = s_i\delta + (2 - g_i)\beta.} \tag{4.5}$$

For an internal excursion, applying Theorem 4.2 across the preceding gap, the excursion, and the following gap gives $g_i \le 1$. The two adjacent copies of (4.4) give $\ell_i, r_i < \beta$; if $g_i \le 0$, then (4.5) would give $\ell_i + r_i > 2\beta$. Thus

$$\boxed{g_i = 1 \quad (2 \le i \le N-1).} \tag{4.6}$$

If there are at least two excursions, the same discrepancy calculation from the first start to the second start, and symmetrically at the right end, gives

$$g_1, g_N \le 2. \tag{4.7}$$

If there is only one excursion, (4.3) applied to its two endpoints gives $g_1 \le 3$.

Consequently any configuration with excess $q \ge 4$ contains, after deleting any nonpositive global endpoint excursions, a consecutive positive branch of total gain $G \ge q$. Its internal gains are all 1 and its endpoint gains are 1 or 2. The remainder of the upper-bound proof eliminates the endpoint value 2 and then treats arbitrarily long gain-one excursions.

5 Endpoint defects and sparse-to-symmetric domination

5.1 Cut contraction

Along a fixed matching diagonal, let θ be the B-edge normal angle and let $\eta(\theta)$ be the matched A-vertex radial angle. Then $\eta' = \alpha/\beta$. The condition that this A-vertex lie strictly outside the matched B half-plane is

$$z \cdot u(\theta) < U(\theta), \qquad U(\theta) = R_n \cos(\eta(\theta) - \theta) - \rho_m.$$

Whenever $\delta < \beta$,

$$U'' + U = R_n \left(1 - \frac{\delta^2}{\beta^2}\right) \cos(\eta - \theta) - \rho_m < R_n - \rho_m < 0.$$

Lemma 5.1 (cut contraction)**.** *Assume $\delta < \beta$. Suppose two genuine cut inequalities on one matching diagonal hold at $\theta_0 < \theta_1$ with $\theta_1 - \theta_0 < \pi$. Then every matched cut inequality at an intermediate angle also holds.*

Proof. Let $H(\theta) = z \cdot u(\theta)$, so $H'' + H = 0$, and put $W = U - H$. The endpoint values of W are positive and $W'' + W < 0$. Subtract the harmonic interpolant with the same endpoint values. If the difference had a negative component (a, b), then its negative $w > 0$ there would satisfy $w'' + w > 0$ with zero endpoint values. Multiplication by w and integration would yield

$$\int_a^b (w')^2 < \int_a^b w^2,$$

contrary to the Dirichlet inequality

$$\int_a^b (w')^2 \ge \left(\frac{\pi}{b-a}\right)^2 \int_a^b w^2 > \int_a^b w^2$$

because $b - a < \pi$. □

5.2 Eliminating a gain-two endpoint

We need one elementary weighted sine inequality.

Lemma 5.2. *Let $R \geq 1$, $w \geq 0$, $0 \leq \xi \leq 1$, and $w + (R+1)\beta < \pi$. Then*

$$2\sin\beta \sum_{k=1}^{R} (k - 1 + \xi)\cos(w + k\beta) < (R + \xi)\sin(w + R\beta). \tag{5.1}$$

Proof. Use $2\sin\beta\cos v = \sin(v+\beta) - \sin(v-\beta)$. If $U = w + (R+1)\beta$, right minus left becomes

$$2\sum_{k=1}^{R} \sin(w + k\beta) + (\xi - 1)\sin(w + \beta) + \xi \sin w - (R - 1 + \xi)\sin U.$$

This is affine in ξ. Since $\sin t \geq (t/U)\sin U$ for $0 \leq t \leq U$ by concavity, the value at $\xi = 0$ is at least $R(w+\beta)\sin U/U > 0$, while the value at $\xi = 1$ is at least $(R+1)w\sin U/U \geq 0$; if $w = 0$, strict concavity makes the latter inequality strict. Hence the expression is positive throughout $[0, 1]$. □

Consider a left endpoint excursion of gain 2, ending at $A_i \cap B_j$. Put that crossing at the origin, take the B_j normal angle to be 0, and let the A_i normal angle be $\gamma > 0$. Let $x, y \in (0, 1)$ be the remaining edge lengths from the crossing to the forward endpoints v_{i+1}, w_{j+1}.

Suppose a later start crossing occurs at (A_{i+h-1}, B_{j+h}) with $h \geq 2$. Its start angle is

$$\gamma + (h-1)\delta - \beta < 0,$$

so

$$\gamma < \beta - (h-1)\delta \leq \beta - \delta. \tag{5.2}$$

In particular $\delta < \beta$, and the two genuine cuts supplied by the endpoint core and this later start lie on the same matching diagonal, at least two levels apart. Lemma 5.1 is therefore applicable and gives the level-2 cut. Direct projection of the matched vertex against the B_{j+2} support yields

$$F := x\sin(2\beta - \gamma) - y\sin 2\beta - \sin\beta - \sin(\alpha - 2\beta + \gamma) > 0. \tag{5.3}$$

Moreover (5.2) gives $2\beta - \alpha - \gamma = \beta - \delta - \gamma > 0$, so the denominator hidden in this projection is positive; no sign reversal is being suppressed.

Put

$$R = \left\lceil \frac{\gamma}{\delta} \right\rceil, \qquad w = R\delta - \gamma \in [0, \delta), \qquad \xi = 1 - \frac{w}{\delta} \in (0, 1],$$

$$A_R = \sin(R\alpha), \qquad B_R = \sin(R\alpha - \gamma),$$

and

$$E_R = \sum_{k=1}^{R} [\sin(k\alpha) - \sin(k\beta + w)].$$

The gain-two core itself gives the strict inside margin

$$D := E_R - xA_R + yB_R > 0. \tag{5.4}$$

Define

$$\kappa = \frac{B_R}{\sin 2\beta}, \qquad P = \frac{\sin\gamma\,\sin(R\alpha + 2\beta - \gamma)}{\sin 2\beta},$$

$$C_* = \sin\beta - \sin(\beta - \delta - \gamma).$$

The elimination identity can be checked coefficient by coefficient. Since $\alpha = \beta + \delta$,

$$\alpha - 2\beta + \gamma = -(\beta - \delta - \gamma),$$

so the constant part of $\kappa F + \kappa C_*$ cancels exactly. The y-coefficient is $-\kappa \sin 2\beta = -B_R$, and the x-coefficient follows from

$$\sin(R\alpha)\sin 2\beta = \sin(R\alpha - \gamma)\sin(2\beta - \gamma) + \sin\gamma\,\sin(R\alpha + 2\beta - \gamma).$$

Consequently

$$\boxed{-D = \kappa F + xP + (\kappa C_* - E_R).} \tag{5.5}$$

All displayed coefficients are positive. The level used in (5.4) lies inside the gain-two core, while the inferred level-2 cut lies later in the same active window; hence their B-index span gives $(R+2)\beta < \pi$. For the only nonobvious sine, put $\vartheta = R\beta + w = R\alpha - \gamma$. If $\delta \le \beta/2$, this integer inequality gives $\pi - (R+2)\beta \ge \beta/2 > w$ (because $m/2$ is an integer or a half-integer); if $\delta > \beta/2$, then $R = 1$ and $\beta < 2\alpha/3$, so, since $n \ge 6$, $\vartheta + 2\beta < \alpha + 2\beta < 7\alpha/3 \le 7\pi/9$. Thus $P > 0$.

The summand in E_R is the integral of cos over an interval of length $(k - 1 + \xi)\delta$. Concavity of sin gives

$$E_R \le \delta \sum_{k=1}^{R} (k - 1 + \xi)\cos(w + k\beta).$$

Moreover $\beta - \delta - \gamma > 0$, so

$$C_* = \int_{\beta-\delta-\gamma}^{\beta} \cos u\, du > (\delta + \gamma)\cos\beta = \delta(R + \xi)\cos\beta.$$

Since $0 \le w < \delta < \beta$ and $(R+2)\beta < \pi$, we also have $w + (R+1)\beta < \pi$, so all hypotheses of Lemma 5.2 are satisfied. That lemma gives $\kappa C_* > E_R$. Equation (5.5), together with $F > 0$, would force $D < 0$, contradicting (5.4).

Hence a gain-two endpoint cannot be followed by two positive gain-one excursions. The reflected statement holds at the right endpoint. The same argument excludes $C + (g = 1) + C$, because the start cut of the right core supplies the required later genuine cut.

It remains to exclude two gain-two cores directly adjacent. If the complementary A-gap between them had increment $a \ge 1$, the right-core start would already give $h = a + 1 \ge 2$ and the preceding argument would apply. Thus necessarily $a = 0$. Let the two relative angles be $\gamma > 0$ and $-\phi < 0$; then

$$\gamma + \phi = \beta.$$

Put

$$R = \left\lceil \frac{\gamma}{\delta} \right\rceil, \quad Q = \left\lceil \frac{\phi}{\delta} \right\rceil, \quad k = R + Q.$$

The active B-window gives $(k+1)\beta < \pi$. We also need $k\alpha \le \pi$. If instead $k\alpha > \pi$, then

$$n \le 2k - 1, \qquad m \ge 2k + 3.$$

Since $n \ge 6$, this implies $k \ge 4$. The ceiling definitions and $\gamma + \phi = \beta$ give $(k-2)\delta < \beta$, equivalently $n > (k-2)(m-n)$. Therefore

$$(2k-1)(k-1) \ge n(k-1) > (k-2)m \ge (k-2)(2k+3),$$

which reduces to $7 > 2k$, contradicting $k \geq 4$. Hence $k\alpha \leq \pi$.

For $0 < \theta < \beta$ and $R = \lceil \theta/\delta \rceil$ with $R\alpha < \pi$, define

$$E_R(\theta) = \sum_{j=1}^{R} [\sin(j\alpha) - \sin(j\beta + R\delta - \theta)].$$

Then

$$\boxed{\frac{E_R(\theta)}{\sin(R\alpha)} < \frac{\theta}{\beta}.} \tag{5.6}$$

Indeed, set $X_j = j\alpha$, $Y_j = j\beta + R\delta - \theta$ and $d_j = X_j - Y_j$. The intervals $[Y_j, X_j]$ lie at the right ends of the cells $[(j-1)\alpha, j\alpha]$, with $0 < d_1 < \cdots < d_R = \theta < \beta$. If

$$c = \frac{\sum_j d_j}{R\alpha} = \frac{\theta - (R-1)\delta/2}{\alpha}$$

and

$$F(t) = |[0,t] \cap \cup_j [Y_j, X_j]| - ct,$$

then $F(j\alpha) = -j(R-j)\delta/2 \leq 0$; F decreases on gaps and increases on selected intervals, so $F \leq 0$ and is not identically zero. Integration by parts gives

$$E_R(\theta) - c\sin(R\alpha) = \int_0^{R\alpha} F(t)\sin t\, dt < 0.$$

Since $c \leq \theta/\alpha < \theta/\beta$, (5.6) follows.

Since $R, Q < k$ and $k\alpha \leq \pi$, both applications satisfy the hypothesis $R\alpha, Q\alpha < \pi$ of (5.6). Applying (5.6) to the two cores gives normalized inside margins $h_R < \gamma/\beta$ and $h_Q < \phi/\beta$, hence $h_R + h_Q < 1$. To record the remaining elimination explicitly, let x, y be the two forward edge remainders at the common middle crossing. After reflecting and reversing the right core, its corresponding remainders are

$$x' = 1 - x + y\frac{\sin\beta}{\sin\phi}, \qquad y' = y\frac{\sin\gamma}{\sin\phi}.$$

Divide the two strict core-margin inequalities by $\sin(R\alpha)$ and $\sin(Q\alpha)$ and add them. The x-terms cancel, and one obtains the necessary condition

$$h_R + h_Q - 1 + yc_1 > 0, \tag{5.7}$$

with

$$c_1 = -\sin\gamma\,[\cot(R\alpha) + \cot(Q\alpha)].$$

Because $R\alpha, Q\alpha \in (0,\pi)$ and $(R+Q)\alpha = k\alpha \leq \pi$,

$$\cot(R\alpha) + \cot(Q\alpha) = \frac{\sin(k\alpha)}{\sin(R\alpha)\sin(Q\alpha)} \geq 0.$$

Thus $c_1 \leq 0$, contradicting $h_R + h_Q < 1$ and (5.7).

We have proved:

Lemma 5.3 (endpoint-defect elimination)**.** *Every positive excursion branch of total gain $G \geq 4$ consists entirely of gain-one excursions; all its internal complementary gaps satisfy (4.4).*

5.3 Sparse gain-one branches

A gain-one excursion need not be a single tooth. Suppose such an excursion starts at (A_i, B_j) and has s interior A-vertices. Then it has $s-1$ interior B-vertices and ends at (A_{i+s}, B_{j+s-1}). Because every complementary gap satisfies $c = a+1$, the starts of all excursions in a pure branch lie on one lifted index diagonal.

Let the branch contain $q \geq 4$ gain-one excursions. From the start-angle increments and (4.4),

$$(q-2)\delta < \beta, \qquad (q-1)\beta < \pi. \tag{5.8}$$

The first inequality follows because the start angles of excursions $2, \ldots, q$ all lie in $(-\beta, 0)$ and consecutive starts advance by $(s+a)\delta \geq \delta$; the second follows from the active B-semicircle. Since $q \geq 4$, the first inequality in (5.8) implies $\delta < \beta$. Hence Lemma 5.1 is applicable to every contraction used in the remainder of this subsection.

We now use only genuine endpoint events. Put $a = \pi/n$, $b = \pi/m$, $\varepsilon = a - b$.

Phase centering. Suppose two symmetric cut constraints occur at half-span $L/2$, where

$$L \geq 3, \qquad L\varepsilon < a, \qquad Lb < \frac{\pi}{2},$$

and the central phase is $|\psi| < b$. Positive sine interpolation of the two upper bounds gives

$$M_L(\psi) = R_n \left[\cos^2\psi \frac{\cos(L\varepsilon)}{\cos(Lb)} - \sin^2\psi \frac{\sin(L\varepsilon)}{\sin(Lb)}\right] - \frac{\rho_m \cos\psi}{\cos(Lb)}. \tag{5.9}$$

For either central outside lower bound used below, let $F = M_L - N$. Writing $x = \cos\psi$ gives

$$F(0) - F(\psi) = (1-x)[A_L(1+x) + B],$$

with

$$A_L = R_n \frac{\sin(La)}{\sin(Lb)\cos(Lb)}, \qquad B \geq R_m - \frac{\rho_m}{\cos(Lb)}.$$

For the even central lower bound this is equality. For the odd one, the coefficient of $\cos\psi$ is $R_m \cos(S\varepsilon)/\cos(Sa)$; since $0 \leq S\varepsilon < Sa < \pi/2$, this coefficient is strictly larger than R_m, which proves the displayed lower bound for B. The bracket is positive. It is enough to prove

$$\frac{\sin(La)}{\sin(Lb)} > \frac{\sin a \cos b}{\sin b(1+\cos b)}. \tag{5.10}$$

The right side is $< a/(2b) < L/[2(L-1)] \leq 3/4$. For the first inequality, after using $\sin a < a$, it is enough to note

$$2b < \tan b + \sin b;$$

the difference vanishes at $b = 0$ and has derivative $\sec^2 b + \cos b - 2 > 0$ for $0 < b < \pi/2$. If $La \leq \pi/2$, the left side exceeds 1; if $La > \pi/2$, then $La < Lb + a < \pi/2 + a$ and, since $n \geq 6$, it exceeds $\cos a \geq \cos(\pi/6) > 3/4$. Thus $F(\psi) > 0$ implies $F(0) > 0$.

Even number of excursions. Let $q = 2r$ and $t = r - 1/2$. By (5.8), $\delta < \beta$. Choose the first outside B-vertex in the central complementary gap. There are r nonempty excursions on each side, so the nearest genuine outer cuts lie at least t diagonal units away. Lemma 5.1 contracts them to the symmetric positions $\pm t$. The central outside vertex has phase $|\psi| < b$. Applying the phase-centering result with $L = q - 1$ yields

$$\boxed{\rho_n - R_m < \frac{R_n \cos(t\delta) - \rho_m}{\cos(t\beta)}.} \tag{5.11}$$

Odd number of excursions. Let $q = 2r + 1 \geq 5$ and let the central excursion have length $S \geq 1$. Its adjacent gap bounds give $S\delta < \beta$. Also $S\alpha < \pi$: otherwise $S \geq n/2$ and $S\delta < \beta$ force $m - n = 1$, while the first-to-last active B-index span is at least $S + q - 2 > m/2$, contradicting the active semicircle.

Take the midpoint of the central excursion as coordinate 0. The two genuine outside events adjacent to it lie at $\pm S/2$. If their midpoint phase is ψ, the start/end crossing orientations give

$$\ell = b + S\varepsilon + \psi, \qquad r = b + S\varepsilon - \psi,$$

so the internal-gap bounds $\ell, r < 2b$ imply

$$|\psi| + S\varepsilon < b.$$

The extreme genuine cuts lie at distance at least

$$T = r + \frac{S-1}{2}.$$

Put $L = 2T = q + S - 2$. By (5.8), $\delta < \beta$, so Lemma 5.1 supplies symmetric outer cuts at $\pm T$. If the midpoint phase is ψ, adding the two central outside inequalities gives

$$N_S(\psi) = \frac{\rho_n - R_m \cos\psi \cos(S\varepsilon)}{\cos(Sa)} < z \cdot u_0. \tag{5.12}$$

The two outer cuts give $z \cdot u_0 < M_L(\psi)$ with M_L from (5.9). From the start of excursion 2 to the start of excursion q, the accumulated index increment is at least $S + q - 3 = L - 1$; both start angles lie in $(-2b, 0)$. Hence $(L-1)\delta < 2b$, equivalently $L\varepsilon < a$. The outer active B-normals are in one strict semicircle, so $L\beta < \pi$, equivalently $Lb < \pi/2$. Together with $|\psi| < b$, phase centering applies.

It remains to show that a long central excursion cannot improve the centered margin. Put

$$c = q - 2, \qquad \eta = b/a, \qquad \lambda = 1 - \eta, \qquad y = Sa, \qquad x = cb + \eta y = Lb,$$

and define

$$\mathcal{G}(y) = [R_n \cos(c\varepsilon + \lambda y) - \rho_m] \frac{\cos y}{\cos x} - \rho_n + R_m \cos(\lambda y). \tag{5.13}$$

This equals $\cos(Sa)[M_L(0) - N_S(0)]$. Differentiation gives

$$\begin{aligned} \mathcal{G}'(y) = \frac{\cos y}{\cos x}\Big[& -R_n \lambda \sin(c\varepsilon + \lambda y) \\ & + \big(R_n \cos(c\varepsilon + \lambda y) - \rho_m\big)(\eta \tan x - \tan y)\Big] - R_m \lambda \sin(\lambda y). \end{aligned} \tag{5.14}$$

The first and last terms are negative and the middle coefficient is negative by Lemma 4.1. If $S' = y/a$ and $L' = c + S'$, then throughout the continuous contraction from S to 1,

$$\eta > \frac{L'-1}{L'}, \qquad \eta\frac{x}{y} = \eta^2 \frac{L'}{S'} > \frac{(L'-1)^2}{L'(L'-c)} > 1,$$

where the last inequality is $(c-2)L' + 1 > 0$. Since $\tan t/t$ is strictly increasing on $(0, \pi/2)$, $\eta \tan x > \tan y$. Hence $\mathcal{G}'(y) < 0$. Therefore feasibility at $S \geq 1$ implies feasibility at $S = 1$, which is exactly

$$\boxed{\frac{\rho_n - R_m \cos(\delta/2)}{\cos(\alpha/2)} < \frac{R_n \cos(r\delta) - \rho_m}{\cos(r\beta)}.} \tag{5.15}$$

Thus every excess-$q \geq 4$ configuration forces a symmetric threshold of length at least q: first pass to a positive branch of gain $G \geq q$, eliminate endpoint defects, and apply (5.11) or (5.15) to its G gain-one excursions.

6 Global realization and the exact trigonometric formula

The preceding argument gives necessary symmetric thresholds. We now prove that they are sufficient by constructing globally valid configurations; this step is separate from the local cut-contraction argument.

Assume first

$$6 \le n < m < 2n.$$

Write $a = \pi/n$, $b = \pi/m$, $\varepsilon = a - b$ and retain $\alpha = 2a$, $\beta = 2b$, $\delta = 2\varepsilon$.

Define

$$H_e = R_m - \rho_n, \qquad H_o = \rho_m - R_n + \frac{1}{2}\tan a,$$

$$\Phi_e(t) = R_n \cos(t\delta) + H_e \cos(t\beta) - \rho_m, \tag{6.1}$$

$$\Phi_o(t) = R_n \cos(t\delta) + H_o \cos(t\beta) - \rho_m. \tag{6.2}$$

Inequality (5.11) is precisely $\Phi_e(r - 1/2) > 0$, and (5.15) is precisely $\Phi_o(r) > 0$.

6.1 The two centered contact configurations

For the even contact configuration, take the A-edge normals to be $u(\alpha i)$, the B-vertex radial directions to be $u(\beta j)$, and put the B-center at

$$h_e e_1, \qquad h_e = \rho_n - R_m = -H_e.$$

The central B-vertex touches the midpoint of the central A-edge.

For the odd contact configuration, take the A-vertex radial directions to be $u(\alpha i)$ and the B-edge normals to be $u(\beta j)$. Put

$$s_* = R_n - \frac{1}{2}\tan a, \qquad h_o = s_* - \rho_m = -H_o.$$

Then the endpoints $(s_*, \pm 1/2)$ of the central B-edge lie on the two A-edges adjacent to the eastern A-vertex, strictly inside those edges.

We must show that, apart from these prescribed contacts, every B-vertex is outside A.

For $h = h_e$ or h_o, set

$$W_h(x) = h\cos(\alpha x) + R_m \cos(\delta x) - \rho_n.$$

A positive value means that the B-vertex indexed by x is excluded by the matched A support. Put $d = m - n$ and $\lambda = d/m < 1/2$, so $\delta = \lambda\alpha$. Then

$$W_h'' + \alpha^2 W_h = \alpha^2 f(x), \qquad f(x) = R_m(1 - \lambda^2)\cos(\delta x) - \rho_n. \tag{6.3}$$

Let $X_0 = n/(2d) + 1$. For $|x| \le X_0$,

$$|\delta x| \le (1 + \lambda)a.$$

Since ρ_t/t is strictly increasing (write $\rho_t/t = [x\cot x]/(2\pi)$ with $x = \pi/t$),

$$R_m(1 - \lambda^2)\cos(\delta x) \ge \rho_m(1 - \lambda^2)\frac{\cos((1+\lambda)a)}{\cos((1-\lambda)a)} > \rho_n(1 + \lambda)\frac{\cos((1+\lambda)a)}{\cos((1-\lambda)a)}.$$

The last factor is > 1. Indeed, for

$$g(\lambda) = (1 + \lambda)\frac{\cos((1+\lambda)a)}{\cos((1-\lambda)a)},$$

we have $g(0) = 1$ and

$$(\log g)' = \frac{1}{1+\lambda} - a\big[\tan((1+\lambda)a) + \tan((1-\lambda)a)\big].$$

For fixed a, the tangent sum is increasing in λ, because its derivative is

$$a\big[\sec^2((1+\lambda)a) - \sec^2((1-\lambda)a)\big] \geq 0;$$

for fixed λ it is increasing in a. Hence, using $a \leq \pi/6$ and $0 \leq \lambda < 1/2$,

$$\tan((1+\lambda)a) + \tan((1-\lambda)a) < \tan\frac{\pi}{4} + \tan\frac{\pi}{12} = 3 - \sqrt{3},$$

and $1/(1+\lambda) > 2/3$. Therefore

$$(\log g)' > \frac{2}{3} - \frac{\pi}{6}(3-\sqrt{3}) > 0;$$

for example $\pi < 22/7$ and $\sqrt{3} > 19/11$ verify the last positivity by exact rational comparison. Thus $f(x) > 0$ on this range.

Put $X = \min(X_0, n/2)$. In the even configuration, $W(0) = W'(0) = 0$, and variation of constants gives

$$W(x) = \alpha \int_0^x \sin(\alpha(x-u)) f(u)\, du > 0 \qquad (0 < x \leq X).$$

In the odd configuration, W is even and $W(\pm 1/2) = 0$. Since $W'(0) = 0$, variation of constants at $x = 1/2$ gives

$$0 = W(0)\cos a + \alpha \int_0^{1/2} \sin(\alpha(1/2-u)) f(u)\, du,$$

so $W(0) < 0$. Differentiating the same formula shows $W'(1/2) > 0$ (both resulting terms are positive, since $a \leq \pi/6$). Restarting the variation-of-constants formula at $1/2$ then gives $W(x) > 0$ for $1/2 < x \leq X$, because $\alpha(x - 1/2) < \pi$. In particular $W(0) = H_e - H_o < 0$, so $H_o > H_e > 0$.

For the remaining vertices, write

$$v(x) = -He_1 + R_m u(\beta x), \qquad H = H_e \text{ or } H_o.$$

On $0 \leq x \leq m/2$, $|v(x)|$ is strictly increasing. Moreover

$$R_m > \frac{m}{2\pi}, \qquad H \geq H_e = R_m - \rho_n > \frac{d}{2\pi},$$

the latter because

$$\rho_t' = \frac{\pi}{2t^2}\csc^2(\pi/t) > \frac{1}{2\pi}.$$

If $x_* = 0$ in the even case and $x_* = 1/2$ in the odd case, then

$$|v(X)|^2 - |v(x_*)|^2 = 4R_m H\big[\sin^2(Xb) - \sin^2(x_* b)\big] > \frac{d}{\pi^2 m}\left(4X^2 - \frac{\pi^2}{4}\right) > \frac{1}{4}. \tag{6.4}$$

For $X = X_0$ the last inequality follows from $4(n/d + 2) > 12 > \pi^2$; for $X = n/2$ it follows from $\pi^2 < 10$ and $d(4n^2 - 20) - 10n > 0$ for $n \geq 6$. Since $R_n^2 - \rho_n^2 = 1/4$, every vertex beyond X lies outside the A circumcircle. Hence all nonprescribed B-vertices are strictly outside A.

The A-center lies in $\operatorname{int} B$ in both contact configurations. Move the B-center a sufficiently small distance in the positive e_1 direction. The prescribed contacts become strict outside vertices while all other strict inequalities remain strict.

Lemma 6.1 (unit-edge cap lemma)**.** *Let $n \geq 4$. Suppose the A-center lies in* $\operatorname{int} B$*, every B-vertex lies outside A, and the two polygons are in general position. Then every B-inside excursion cuts off exactly one A-vertex, so every A-edge retains a positive-length part.*

Proof. An excursion is a strict subsegment of a single unit B-edge, hence has length < 1. The cut-off A-cap does not contain the center. If it contained at least two A-vertices, its two crossing points would lie on nonadjacent unit A-edges, whose distance is at least 1 by Lemma 3.1, impossible. If it contained no A-vertex, two noncollinear line segments would intersect twice, also impossible. Thus exactly one vertex is cut off. Maximal outside arcs of ∂A alternate with the B-inside excursions, so every such outside arc contains exactly that one vertex. Consequently no complete A-edge belongs to an outside arc, and every A-edge retains a positive-length part. □

In the even contact configuration, direct scalar products show that the cut margin at half-integer x is $\Phi_e(x)$; in the odd configuration, the cut margin at integer x is $\Phi_o(x)$. Both coefficients H_e, H_o are positive: $H_e = R_m - \rho_n > 0$, while $H_o > 0$ follows from $\rho_m > R_n$ (Lemma 4.1). Hence for $j \in \{e, o\}$,

$$\Phi_j'(t) = -R_n\delta\sin(t\delta) - H_j\beta\sin(t\beta) < 0 \qquad (0 < t < m/4),$$

because $0 < t\delta < t\beta < \pi/2$. Thus both margin functions are strictly decreasing on the whole range used below. Therefore:

Proposition 6.2 (sufficiency of the symmetric thresholds)**.** *For $6 \leq n < m < 2n$:*

(i) if $q = 2r \geq 4$ and $\Phi_e(r - 1/2) > 0$, there is a configuration with at least $n + q$ sides;

(ii) if $q = 2r + 1 \geq 5$ and $\Phi_o(r) > 0$, there is a configuration with at least $n + q$ sides.

Proof. In the even case the q half-integer positions $-r + 1/2, \ldots, r - 1/2$ remain strict cuts after the small displacement. In the odd case the q integer positions $-r, \ldots, r$ do. All B-vertices are outside A and the A-center remains inside B, so Lemma 6.1 converts these cuts into q distinct one-vertex caps. Hence the intersection has at least $n + q$ sides. □

Combining the upper reduction from Sections 4–5 with Proposition 6.2 gives the exact trigonometric upper/lower match for $n \geq 6$ and $m < 2n$.

6.2 The universal $+3$ construction

We also need the baseline excess 3.

If $n < m < 2n$, the odd margin at $t = 1$ simplifies exactly to

$$\Phi_o(1) = \frac{2\cos a - 1}{2\cos a}\sin(2b - a) > 0. \tag{6.5}$$

For $n \geq 6$, the same odd centered contact construction gives $q = 3$. For $n = 4, 5$, use that odd configuration directly. Its two central B-vertices lie on the two A-edges adjacent to the eastern vertex, and $s_* = R_n - \frac{1}{2}\tan a > 0$, so the A-center lies strictly inside B. The first noncontact B-vertex has vertical coordinate $1/2 + \cos 2b > R_n$ (for $n = 4$ use $1/2 + \cos(2\pi/5) > 3/4 > R_4$; for $n = 5$ use $1/2 + \cos(\pi/3) = 1 > R_5$), and the distances of subsequent vertices from the A-center increase. Hence all noncentral B-vertices are outside A. A small displacement makes the two contact vertices strict as well, and Lemma 6.1 gives $n + 3$ sides.

If $m = 2n$, put $a = \pi/n$. Orient B so its edge normals are separated by a and place its center at $-(R_n/\cos a)e_1$. The identity $\rho_{2n} = R_n + \rho_n$ shows that the central B-edge cuts the eastern

A-vertex, while the two adjacent B supports coincide with the two adjacent A supports. Every other B support is strictly redundant: for odd support index j the support slack is

$$R_n\left(1-\frac{\cos(ja)}{\cos a}\right),$$

vanishing only at $j=\pm 1$, and for nonzero even j it is

$$\frac{R_n}{\cos a}[\cos^2 a-\cos(ja)]>0.$$

Thus the initial intersection has $n+1$ distinct support lines, including two positive-length overlaps. Choose an interior point on each overlap. Rotate B slightly and apply the unique small translation that keeps the two rotated B supports through those two chosen points. The two overlaps split into four positive-length edges while all strict supports persist, giving $n+3$ sides.

If $m>2n$, place the central B-edge with endpoints $(s,\pm 1/2)$ and interior to its left, where s is slightly below

$$s_*=R_n-\frac{1}{2}\tan a.$$

The eastern A-vertex is then outside B, whereas all other A-vertices lie in the B incircle. It suffices to check the two A-vertices adjacent to the eastern one. If $c=\cos a$ and

$$t_*=(c-1/2)\tan a,$$

the critical incircle radius is

$$\rho_*=\frac{t_*^2+c^2}{2t_*},$$

and direct simplification gives

$$\rho_*-\rho_{2n}=\frac{(1-c)^2}{4c(2c-1)\sin a}.$$

For $n=4$ this is $(\sqrt{2}-1)/4<1/8$. For $n\geq 5$, rewrite it as

$$\frac{\sin^3 a}{4c(2c-1)(1+c)^2}<\frac{5}{144}<\frac{1}{8},$$

using $c>4/5$ and $\sin a<3/5$. On the other hand $\rho_x'>1/(2\pi)>1/8$, so $\rho_m\geq\rho_{2n+1}>\rho_*$. Thus $n-1$ vertices of A lie inside B, at least two B-vertices lie inside A, and the boundaries cross twice; Lemma 2.2 gives at least $n+3$ sides.

We have therefore proved

$$\boxed{K(n,m)\geq n+3\qquad(n\geq 4,\ m>n).}\tag{6.6}$$

Together with (3.1) and Lemma 3.4, this completes the two low rows:

$$\boxed{K(4,m)=7\ (m>4),\qquad K(5,m)=8\ (m>5).}\tag{6.7}$$

6.3 Exact trigonometric formula for $n \geq 6$

If $m \geq 2n$, then $\delta \geq \beta$. Any positive branch of gain at least 4 would, after endpoint elimination, contain at least four gain-one excursions, but (5.8) would require $(q-2)\delta < \beta$, impossible. Hence (6.6) gives

$$K(n,m) = n+3 \qquad (n \geq 6,\ m \geq 2n). \tag{6.8}$$

If $n < m < 2n$, define $Q^{\mathrm{adm}}_{\mathrm{trig}}(n,m)$ to be the largest $q \geq 4$ satisfying the angular admissibility conditions and the corresponding strict even or odd threshold (5.11), (5.15). The necessity proved in Section 5 and the sufficiency proved above give

$$\boxed{K(n,m) = n + \max\{3, Q^{\mathrm{adm}}_{\mathrm{trig}}(n,m)\}.} \tag{6.9}$$

This is the exact trigonometric formula that will now be integerized.

7 Integerization of the centered threshold

Set

$$d = m - n \geq 1, \qquad a = \frac{\pi}{n}, \qquad b = \frac{\pi}{m}, \qquad \varepsilon = a - b.$$

Then

$$\frac{\varepsilon}{b} = \frac{d}{n}.$$

For $q \geq 4$ define

$$\boxed{c_q = \left\lfloor \frac{q(q-2)}{2} \right\rfloor.} \tag{7.1}$$

Thus

$$c_{2r} = 2r(r-1), \qquad c_{2r+1} = 2r^2 - 1.$$

Put

$$z = n - c_q d.$$

7.1 Exact sine sums

For $q = 2r$, $s = q-1 = 2r-1$, clearing the positive denominators in the even threshold (5.11) gives

$$S_{2r} = \sin(sb) + 2\sum_{k=1}^{r-1} \sin(sb - 2ka) - 2\sum_{k=1}^{r-1} \sin(2kb) > 0. \tag{7.2}$$

For $q = 2r+1$, $s = 2r$, the odd threshold (5.15) becomes

$$\begin{aligned} S_{2r+1} = {} & \sin(sb) - \sin(s\varepsilon) + 2\sum_{j=1}^{r-1} \sin(2jb - 2(r-j)\varepsilon) \\ & - \sum_{j=1}^{r} \sin((2j-1)b - \varepsilon) - \sum_{j=1}^{r-1} \sin((2j+1)b + \varepsilon) > 0. \end{aligned} \tag{7.3}$$

Let

$$\psi(t) = t - \sin t.$$

In both parities the linear part of the corresponding sine sum is

$$b - c_q\varepsilon = \frac{\varepsilon}{d}(n - c_q d) = \frac{\varepsilon}{d}z.$$

Hence

$$\boxed{S_q = \frac{\varepsilon}{d}z - \Psi_q, \qquad S_q > 0 \iff z > \Theta_q := \frac{d}{\varepsilon}\Psi_q.} \tag{7.4}$$

7.2 Peano kernels

Since $\psi(0) = \psi'(0) = \psi''(0) = 0$ and $\psi'''(t) = \cos t$,

$$\psi(u) = \frac{1}{2}\int_0^u (u-t)^2 \cos t\, dt.$$

Thus

$$\Psi_q = \frac{1}{2}\int K_q(t)\cos t\, dt.$$

For even $q = 2r$,

$$K_{2r}(t) = (sb - t)_+^2 + 2\sum_{k=1}^{r-1}(sb - 2ka - t)_+^2 - 2\sum_{k=1}^{r-1}(2kb - t)_+^2. \tag{7.5}$$

For odd $q = 2r + 1$,

$$\begin{aligned} K_{2r+1}(t) = {} & (sb - t)_+^2 - (s\varepsilon - t)_+^2 \\ & + 2\sum_{j=1}^{r-1}(2jb - 2(r-j)\varepsilon - t)_+^2 \\ & - \sum_{j=1}^{r}((2j-1)b - \varepsilon - t)_+^2 \\ & - \sum_{j=1}^{r-1}((2j+1)b + \varepsilon - t)_+^2. \end{aligned} \tag{7.6}$$

Lemma 7.1 (Peano-kernel positivity)**.** *If*

$$\frac{\varepsilon}{b} \le \frac{1}{c_q},$$

then $K_q(t) \ge 0$ for all t.

Proof. We record the finite computation because it is the key exact step. For $q = 2r$, the knots in (7.5) interlace. The second derivative of K alternates between $+2$ and -2 on successive knot intervals, so only minima in convex intervals need checking. If p pairs of knots remain to the right of such an interval, its minimum is

$$M_p^{\mathrm{e}} = \frac{2pb^2}{3}\left[3 - 4x(p^2-1) - 2x^2(3p^3 + 4p^2 - 1)\right], \quad x = \varepsilon/b.$$

For $p \ge 1$ this bracket is strictly decreasing in both variables: its partial derivatives are

$$-4(p^2-1) - 4x(3p^3 + 4p^2 - 1) < 0, \qquad -8xp - 2x^2(9p^2 + 8p) < 0.$$

Hence its worst value occurs at $p = r-1$, $x = 1/(2r(r-1))$, where it equals

$$\frac{(2r-1)(r^2-r+1)}{2r(r-1)^2} > 0.$$

For $q = 2r+1$, the curvature pattern is $-2, 0, +2, -2, 0, +2, \ldots$. The convex-interval minimum with p pairs remaining is

$$M_p^{\circ} = \frac{2pb^2}{3}\left[3 - 4x(p^2-1) - x^2(6p^3+8p^2+1)\right].$$

Here too the bracket is strictly decreasing for $p \geq 1$, since

$$\partial_x B = -4(p^2-1) - 2x(6p^3+8p^2+1) < 0, \qquad \partial_p B = -8xp - x^2(18p^2+16p) < 0.$$

Therefore the worst case is $p = r-1$, $x = 1/(2r^2-1)$, giving

$$\frac{2r(2r^3+5r^2+r-5)}{(2r^2-1)^2} > 0.$$

Thus all local minima are nonnegative. The kernel is not identically zero: immediately to the left of its largest knot sb, only the term $(sb-t)_+^2$ survives. Hence K_q is positive on a nonempty interval. □

Theorem 7.2 (universal ratio obstruction)**.** *If a centered canonical q-chain with $q \geq 4$ is feasible, then*

$$\boxed{n > dc_q.} \tag{7.7}$$

Proof. Fix b and suppose, toward a contradiction, that a centered canonical q-chain is feasible at some $\varepsilon \geq \varepsilon_0 := b/c_q$. The half-circle admissibility condition $(q-1)\beta < \pi$ is independent of ε, and the normal-fan condition is

$$(q-2)\varepsilon < b.$$

Hence every $\varepsilon' \in [\varepsilon_0, \varepsilon]$ is angularly admissible as well. In particular the support of K_q at ε_0 lies in

$$[0, T], \qquad T = (q-1)b < \frac{\pi}{2}.$$

Lemma 7.1 gives $K_q \geq 0$, and the kernel is positive on a nonempty interval; since $\cos t > 0$ on $[0, T]$, we have $\Psi_q(\varepsilon_0) > 0$. The linear part in (7.4) vanishes at ε_0, so $S_q(\varepsilon_0) < 0$. It remains to prove that S_q is strictly decreasing on the whole interval $[\varepsilon_0, \varepsilon]$.

For even $q = 2r$,

$$\frac{\partial S_{2r}}{\partial \varepsilon} = -4\sum_{k=1}^{r-1} k\cos(sb - 2k(b+\varepsilon)).$$

Every displayed cosine is positive. Indeed the largest argument is $< sb < \pi/2$, while the smallest is

$$sb - 2(r-1)a = b - 2(r-1)\varepsilon > 0,$$

the last inequality being equivalent to $(2r-1)\varepsilon < a$. Thus the derivative is strictly negative.

For odd $q = 2r + 1$, differentiating (7.3) and pairing terms gives

$$\begin{aligned}\frac{\partial S_{2r+1}}{\partial \varepsilon} = & -s\cos(s\varepsilon) - 4\sum_{j=1}^{r-1}(r-j)\cos P_j \\ & + \cos(2rb-a) + 2\sin a\sum_{j=1}^{r-1}\sin(2jb),\end{aligned}$$

where $s = 2r$ and $P_j = 2jb - 2(r-j)\varepsilon$. Here $0 < P_j < \pi/2$: the upper bound follows from $rb < \pi/4$, and the lower bound from $(2r-1)\varepsilon < a$, which implies $2(r-1)\varepsilon < b$. Hence the middle sum is nonpositive and may be discarded in an upper bound.

Since the standing scope is $n \geq 6$, we have $a \leq \pi/6$; together with $s\varepsilon < a$ this gives

$$\cos(s\varepsilon) > \cos(\pi/6) = \frac{\sqrt{3}}{2} > \frac{6}{7}.$$

Also, using $\sin x < x$, $rb < \pi/4$, and

$$2r\varepsilon < a = b + \varepsilon \quad \Longrightarrow \quad a < \frac{2r}{2r-1}b,$$

we obtain

$$\begin{aligned}2\sin a\sum_{j=1}^{r-1}\sin(2jb) &< 2a\sum_{j=1}^{r-1}2jb = 2ab\,r(r-1) \\ &< \frac{4r^2(r-1)}{2r-1}b^2 < \frac{\pi^2(r-1)}{4(2r-1)} < \frac{\pi^2}{8} < \frac{5}{4} < \frac{11}{7}.\end{aligned}$$

Finally $\cos(2rb - a) \leq 1$. Since $s \geq 4$, the derivative is therefore

$$< -4\cdot\frac{6}{7} + 1 + \frac{11}{7} = -\frac{6}{7} < 0.$$

Thus $S_q(\varepsilon) < S_q(\varepsilon_0) < 0$, contradicting feasibility. Hence no feasible chain can satisfy $\varepsilon \geq b/c_q$. Since $\varepsilon/b = d/n$, this is (7.7). □

7.3 Cubic shadows

Define the cubic shadow of the Peano-kernel correction by

$$C_q := \frac{1}{2}\int_0^\infty K_q(t)\,dt, \qquad \Theta_q^{(3)} := \frac{d}{\varepsilon}C_q. \tag{7.8}$$

Whenever $\varepsilon/b \leq 1/c_q$, Lemma 7.1 gives $K_q \geq 0$, while $\cos t \leq 1$; hence

$$\Theta_q \leq \Theta_q^{(3)}. \tag{7.9}$$

Put

$$c = c_q, \qquad x = \frac{\varepsilon}{b} = \frac{d}{n}.$$

Since $\int_0^u (u-t)^2\,dt = u^3/3$ and, with $x = \varepsilon/b = d/n$,

$$b = \frac{\pi x}{d(1+x)},$$

direct integration of the positive-part quadratics in (7.5)–(7.6), followed by collecting the signed cubic knot sums, gives

$$\Theta_{2r}^{(3)} = \frac{\pi^2}{6d} f_{\mathrm{e}}(c,x), \qquad f_{\mathrm{e}}(c,x) = \frac{xP_{\mathrm{e}}(c,x)}{(1+x)^2}, \tag{7.10}$$

where

$$P_{\mathrm{e}}(c,x) = 3c + 1 - c(c-1)x + c(c+2)x^2 - c^2x^3, \tag{7.11}$$

and

$$\Theta_{2r+1}^{(3)} = \frac{\pi^2}{6d} f_{\mathrm{o}}(c,x), \qquad f_{\mathrm{o}}(c,x) = \frac{xP_{\mathrm{o}}(c,x)}{(1+x)^2}, \tag{7.12}$$

with

$$P_{\mathrm{o}}(c,x) = 3c + 4 - (c^2-4)x + (c^2-3c-1)x^2 - (c^2+4c+2)x^3. \tag{7.13}$$

The lower-tail argument below needs monotonicity in x, so we record it explicitly rather than infer it from an endpoint maximum.

Lemma 7.3 (monotonicity of the cubic shadows)**.** *For every $c > 1$, $f_{\mathrm{e}}(c,x)$ is strictly increasing on $0 \le x \le 1/c$. For every $c \ge 7$, $f_{\mathrm{o}}(c,x)$ is strictly increasing on the same interval.*

Proof. Write

$$(1+x)^3 \frac{\partial f_{\mathrm{e}}}{\partial x} = N_{\mathrm{e}}(c,x).$$

For $y = cx \in [0,1]$, exact expansion gives

$$\begin{aligned} c^2 N_{\mathrm{e}}(c, y/c) = {} & c^3(3-2y) + c^2(3y^2 - y + 1) \\ & + cy(-3y^2 + 6y - 1) + 2y^3(1-y). \end{aligned} \tag{7.14}$$

On $[0,1]$, the first term is at least c^3; the quadratic $3y^2 - y + 1$ is positive because its discriminant is -11; moreover $-3y^2 + 6y - 1 \ge -1$, and the last term is nonnegative. Therefore

$$c^2 N_{\mathrm{e}}(c, y/c) > c^3 - c > 0 \qquad (c > 1).$$

Thus f_{e} is strictly increasing.

Likewise, if

$$(1+x)^3 \frac{\partial f_{\mathrm{o}}}{\partial x} = N_{\mathrm{o}}(c,x),$$

then

$$\begin{aligned} c^4 N_{\mathrm{o}}(c, y/c) = {} & c^5(3-2y) + c^4(3y^2 - 3y + 4) \\ & + c^3(-3y^3 - 9y^2 + 4y) \\ & - c^2(2y^4 + 19y^3 + 3y^2) - c(8y^4 + 9y^3) - 4y^4. \end{aligned} \tag{7.15}$$

For $0 \le y \le 1$, this is bounded below by

$$c^5 - 12c^3 - 24c^2 - 17c - 4.$$

If $c \ge 7$, then

$$c^5 - 12c^3 - 24c^2 - 17c - 4 \ge 37c^3 - 24c^2 - 17c - 4 > 0,$$

because $c^2 - 12 \ge 37$ and, for $c \ge 7$, $37c^3 - 24c^2 - 17c - 4 = c^2(37c - 24) - 17c - 4 \ge 235c^2 - 17c - 4 > 0$. Hence f_{o} is strictly increasing for $c \ge 7$. □

Since even $q \geq 4$ has $c \geq 4$, Lemma 7.3 gives

$$f_{\mathrm{e}}(c, x) \leq f_{\mathrm{e}}(c, 1/c) = \frac{2c+1}{c+1} < 2. \tag{7.16}$$

For odd $q \geq 5$, $c \geq 7$, and similarly

$$f_{\mathrm{o}}(c, x) \leq f_{\mathrm{o}}(c, 1/c) = \frac{(c-1)(c+2)(2c+1)}{c^2(c+1)}.$$

Moreover

$$\frac{21}{10} - f_{\mathrm{o}}(c, 1/c) = \frac{c^3 - 9c^2 + 30c + 20}{10c^2(c+1)} > 0 \qquad (c \geq 7),$$

because the numerator is positive at $c = 7$ and has derivative $3((c-3)^2 + 1) > 0$. Consequently

$$\boxed{\Theta_{2r} < \frac{\pi^2}{3d}, \qquad \Theta_{2r+1} < \frac{7\pi^2}{20d}.} \tag{7.17}$$

These inequalities already settle all $d \geq 4$.

Corollary 7.4. *If $d \geq 4$ and $q \geq 4$, then a centered canonical q-chain is feasible if and only if*

$$n > dc_q.$$

Proof. Necessity is Theorem 7.2. Conversely, if $n > dc_q$ then $z \geq 1$. By (7.17), $\Theta_q < 1$ for $d \geq 4$, so (7.4) gives $S_q > 0$. The angular conditions also hold: since $c_q \geq q - 2$, one has $n > (q-2)d$ and therefore $m = n + d > (q-1)d$, which is equivalent to $(q-1)\delta < \alpha$; moreover

$$m \geq d(c_q + 1) + 1 > 2(q-1)$$

because $d \geq 4$ and $c_q \geq q - 2$. Thus the centered chain is genuinely admissible. □

8 The finite curvature corrections

Lemma 8.1 (rational bounds for π)**.**

$$\frac{333}{106} < \pi < \frac{355}{113}.$$

Proof. Machin's identity

$$\frac{\pi}{4} = 4 \arctan \frac{1}{5} - \arctan \frac{1}{239}$$

follows from the tangent addition formula. For $0 < x < 1$ the alternating series

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$$

has remainder bounded in magnitude by the first omitted term. Taking four terms for $x = 1/5$ and one term for $x = 1/239$, and performing the resulting rational comparisons, gives the displayed bounds. In particular $493/50 < \pi^2 < 10$ and $\pi < 22/7$. □

For $d = 1, 2, 3$ we also need a lower cubic shadow. Assume a candidate chain is feasible. Theorem 7.2 gives $n > dc_q$, hence $x = d/n < 1/c_q$; Lemma 7.1 therefore gives $K_q \geq 0$. Its angular admissibility gives

$$T := (q-1)b < \frac{\pi}{2}.$$

The support of K_q is contained in $[0, T]$. Since $\cos T \leq \cos t \leq 1$ on this interval,

$$\boxed{\cos T\, \Theta_q^{(3)} \leq \Theta_q \leq \Theta_q^{(3)}.} \tag{8.1}$$

This is the only lower-shadow estimate used below. We use $493/50 < \pi^2 < 10$ from Lemma 8.1 and $\cos u > 1 - u^2/2$ for $0 < u < \pi/2$. Whenever a bound below replaces s by the first value in a tail, we use the elementary fact that $s/(s^2 + A)$ is strictly decreasing for $s^2 > A$, since its derivative is $(A - s^2)/(s^2 + A)^2$.

The case $d = 2$. By (7.17), every $z \geq 2$ satisfies the strict sine threshold $S_q > 0$. It remains to exclude $z = 1$ except at the finite boundary cases listed below.

For odd $q \geq 5$, put $s = q - 1 \geq 4$. Then $c = (s^2 - 2)/2$, $x = 2/(s^2 - 1)$, and direct substitution gives

$$f_{\mathrm{o}}(c, x) - 2 = \frac{s^2(s^4 - 16s^2 + 11)}{(s-1)^2(s+1)^2(s^2+1)^2} > 0. \tag{8.2}$$

Indeed, with $t = s^2 \geq 16$, $t^2 - 16t + 11 = t(t - 16) + 11 > 0$. Moreover

$$T = \frac{s\pi}{s^2+1}, \qquad T^2 < \frac{160}{289}, \qquad \cos T > \frac{209}{289} > \frac{7}{10}.$$

Hence (8.1) gives the strict rational lower bound

$$\Theta_q > \frac{7}{10}\frac{493}{50}\frac{2}{12} = \frac{3451}{3000} > 1. \tag{8.3}$$

For even $q \geq 6$, $s = q - 1 \geq 5$, $c = (s^2 - 1)/2$, and $x = 2/s^2$. Here

$$f_{\mathrm{e}}(c, x) - \frac{9}{5} = \frac{s^8 - 11s^6 - 51s^4 + 10s^2 - 20}{5s^4(s^2+2)^2} > 0. \tag{8.4}$$

For $s \geq 5$, the numerator is at least

$$s^4(14s^2 - 51) - 20 > 0,$$

because $s^2 - 11 \geq 14$. Also

$$T = \frac{s\pi}{s^2+2}, \qquad T^2 < \frac{250}{729}, \qquad \cos T > \frac{604}{729} > \frac{4}{5}.$$

Thus

$$\Theta_q > \frac{4}{5}\frac{493}{50}\frac{1}{12}\frac{9}{5} = \frac{1479}{1250} > 1. \tag{8.5}$$

The only $d = 2$ boundary not covered analytically is $q = 4, z = 1$.

The case $d = 3$. Again (7.17) gives $S_q > 0$ for every $z \geq 2$. At $z = 1$, for odd $q \geq 7$ put $s = q - 1 \geq 6$. Then

$$f_{\mathrm{o}}(c, x) - 2 = \frac{4(27s^6 - 333s^4 + 222s^2 - 20)}{(3s^2 - 4)^2(3s^2 + 2)^2} > 0. \tag{8.6}$$

Indeed

$$27s^6 - 333s^4 + 222s^2 - 20 = s^4(27s^2 - 333) + 222s^2 - 20 > 0$$

for $s \geq 6$. Furthermore

$$T = \frac{2s\pi}{3s^2 + 2}, \qquad T^2 < \frac{72}{605}, \qquad \cos T > \frac{569}{605} > \frac{47}{50},$$

so

$$\Theta_q > \frac{47}{50}\frac{493}{50}\frac{2}{18} = \frac{23171}{22500} > 1. \tag{8.7}$$

For even $q \geq 8$, $s = q - 1 \geq 7$,

$$f_{\mathrm{e}}(c, x) - \frac{97}{50} = \frac{243s^8 - 10152s^6 - 23238s^4 + 22440s^2 - 11725}{50(3s^2 - 1)^2(3s^2 + 5)^2} > 0. \tag{8.8}$$

Since $s^2 \geq 49$, the numerator is larger than

$$s^4(1755s^2 - 23238) - 11725 > 0.$$

Also

$$T = \frac{2s\pi}{3s^2 + 5}, \qquad T^2 < \frac{1960}{23104}, \qquad \cos T > 1 - \frac{980}{23104} > \frac{19}{20}.$$

Consequently

$$\Theta_q > \frac{19}{20}\frac{493}{50}\frac{1}{18}\frac{97}{50} = \frac{908599}{900000} > 1. \tag{8.9}$$

Only $q = 4, 5, 6$ remain for $d = 3, z = 1$.

The case $d = 1$. The upper shadow (7.17) gives $S_q > 0$ for every $z \geq 4$. To rule out the lower layers on the infinite tail, assume $z \leq 3$. Theorem 7.2 then gives $1 \leq z \leq 3$.

For even $q \geq 18$, let $s = q - 1 \geq 17$ and $c = (s^2 - 1)/2$. Since

$$x = \frac{1}{c + z} \geq \frac{1}{c + 3},$$

Lemma 7.3 gives

$$f_{\mathrm{e}}(c, x) \geq f_{\mathrm{e}}\left(c, \frac{1}{c + 3}\right).$$

Direct substitution yields

$$f_{\mathrm{e}}\left(c, \frac{1}{c + 3}\right) - \frac{197}{100} = \frac{P_{\mathrm{e},1}(s)}{100(s^2 + 5)^2(s^2 + 7)^2}, \tag{8.10}$$

where

$$P_{\mathrm{e},1}(s) = 3s^8 - 728s^6 - 18958s^4 - 126280s^2 - 264725.$$

For $s^2 \geq 289$,

$$\begin{aligned} P_{\mathrm{e},1}(s) &\geq 139s^6 - 18958s^4 - 126280s^2 - 264725 \\ &= s^4(139s^2 - 18958) - 126280s^2 - 264725 \\ &\geq 21213 \cdot 289\, s^2 - 126280s^2 - 264725 > 0. \end{aligned} \tag{8.11}$$

Moreover, since $m = c + z + 1 \geq c + 2$,

$$T \leq \frac{2s\pi}{s^2+3}, \qquad T^2 < \frac{11560}{85264}, \qquad \cos T > \frac{93}{100}.$$

Hence

$$\Theta_q > \frac{93}{100}\frac{493}{50}\frac{1}{6}\frac{197}{100} = \frac{3010751}{1000000} > 3. \tag{8.12}$$

For odd $q \geq 17$, let $s = q-1 \geq 16$ and $c = (s^2-2)/2$. Again $x \geq 1/(c+3)$, and Lemma 7.3 gives

$$f_{\mathrm{o}}(c,x) \geq f_{\mathrm{o}}\left(c, \frac{1}{c+3}\right).$$

Here

$$f_{\mathrm{o}}\left(c, \frac{1}{c+3}\right) - \frac{99}{50} = \frac{P_{\mathrm{o},1}(s)}{50(s^2+4)^2(s^2+6)^2}, \tag{8.13}$$

where

$$P_{\mathrm{o},1}(s) = s^8 - 180s^6 - 5652s^4 - 28720s^2 - 35424.$$

For $s^2 \geq 256$,

$$\begin{aligned} P_{\mathrm{o},1}(s) &\geq 76s^6 - 5652s^4 - 28720s^2 - 35424 \\ &= s^4(76s^2 - 5652) - 28720s^2 - 35424 \\ &\geq 13804 \cdot 256\, s^2 - 28720s^2 - 35424 > 0. \end{aligned} \tag{8.14}$$

Also

$$T \leq \frac{2s\pi}{s^2+2}, \qquad T^2 < \frac{10240}{66564}, \qquad \cos T > \frac{923}{1000}.$$

Therefore

$$\Theta_q > \frac{923}{1000}\frac{493}{50}\frac{1}{6}\frac{99}{50} = \frac{15016287}{5000000} > 3. \tag{8.15}$$

Thus all $d = 1$ cases with $q \geq 17$ are decided analytically, and only $4 \leq q \leq 16$ require finite checks within the standing scope $n \geq 6$.

All finite signs not decided by the analytic tails in the range $n \geq 6$ are certified by the accompanying script `integer_staircase_certificate.py`. There is one point at which the bookkeeping matters. For $d = 1, q = 4$ the asserted threshold is $z = 2$, whereas the analytic upper-shadow argument proves feasibility only for $z \geq 4$. Thus both $z = 2$ and $z = 3$ must be checked separately. The certificate does exactly this: it certifies

$$S_4(6,7) > 0, \qquad S_4(7,8) > 0,$$

so no monotonicity in z is being used. The layer $z = 1$, namely $(n,m) = (5,6)$, is intentionally outside both the certificate and Theorem 8.2, because the complete row $n = 5$ was proved independently in Section 6.

More generally, the script contains an internal coverage audit. For $d = 1$ and $4 \leq q \leq 16$ it requires every unresolved layer $z \in \{1,2,3\}$ with $n \geq 6$, with the sign prescribed by the displayed threshold; when the threshold itself is $z = 4$ it also retains the positive $z = 4$ check as a redundant cross-audit. Together with the four small $d = 2,3$ boundary cases, this gives 49 certified signs. The program aborts if any required layer is missing, duplicated, assigned the wrong expected sign, or has an interval meeting zero.

Every theorem-level arithmetic operation in the program uses exact `Fraction` arithmetic. It derives its own rational enclosure of π from Machin's identity and alternating arctangent series, then encloses every sine term in (7.2)–(7.3) by alternating Taylor sums. There is no floating-point sign decision.

The certified boundary signs yield the following theorem.

Theorem 8.2 (integer chain threshold)**.** *Fix integers $n \geq 6$, $d \geq 1$, and $q \geq 4$, and put $m = n + d$. A centered canonical q-chain for the pair (n, m) is feasible if and only if*

$$n \geq n_{\min}(d, q) := dc_q + \kappa_d(q),$$

where

$$\kappa_d(q) = \begin{cases} 1, & d \geq 4, \\ 1, & d = 3,\ q = 4, \\ 2, & d = 3,\ q \geq 5, \\ 2, & d = 2, \\ 2, & d = 1,\ q = 4, \\ 3, & d = 1,\ q \in \{5, 6, 7, 8, 10\}, \\ 4, & d = 1,\ q = 9 \text{ or } q \geq 11. \end{cases} \tag{8.16}$$

Proof. Theorem 7.2 first gives the universal necessity $z = n - dc_q \geq 1$. For $d \geq 4$, Corollary 7.4 already proves that $z \geq 1$ is necessary and sufficient, giving $\kappa_d(q) = 1$.

For $d = 2$, (7.17) gives the strict sine inequality $S_q > 0$ whenever $z \geq 2$. Equations (8.2)–(8.5) exclude $z = 1$ for all $q \geq 5$, and the certificate excludes the remaining case $q = 4, z = 1$. Hence the sine threshold is exactly $z \geq 2$.

For $d = 3$, (7.17) likewise gives $S_q > 0$ for every $z \geq 2$. Equations (8.6)–(8.9) exclude $z = 1$ for $q \geq 7$; the exact certificate gives $z = 1$ positive for $q = 4$ and negative for $q = 5, 6$. This is exactly the stated $d = 3$ row.

For $d = 1$, (7.17) gives $S_q > 0$ for every $z \geq 4$, while (8.10)–(8.15) exclude every $z \leq 3$ once $q \geq 17$. The certificate checks every unresolved layer $z \in \{1, 2, 3\}$ for $4 \leq q \leq 16$, subject only to $n \geq 6$, and its mechanically audited sign ledger is precisely the $d = 1$ row displayed above.

It remains only to verify the two angular admissibility conditions at every claimed positive threshold. Since $c_q \geq q - 2$ and every displayed $\kappa_d(q)$ is positive,

$$n \geq dc_q + \kappa_d(q) > (q - 2)d,$$

so $m = n + d > (q - 1)d$, which is equivalent to $(q - 1)\delta < \alpha$. For the half-circle condition, if $d = 1, q = 4$ then the threshold gives $m \geq 7 > 6 = 2(q - 1)$. If $q \geq 5$, then $c_q > 2q - 4$, and hence already for $d = 1$

$$m \geq c_q + \kappa_1(q) + 1 > 2(q - 1);$$

for $d \geq 2$ the left side only increases. Thus $(q - 1)\beta < \pi$ as well. The strict sine threshold and both angular conditions are therefore equivalent to the stated integer threshold in every case. □

9 The exact formula

Theorem 9.1 (complete formula for $K(n, m)$)**.** *Let $n, m \geq 3$ and put $N = \min\{n, m\}$, $M = \max\{n, m\}$. Then:*

(a) If $N = M$, then $K(n, m) = 2N$.

(b) If $N = 3 < M$, then $K(n, m) = 5$.

(c) If $N = 4 < M$, then $K(n, m) = 7$.

(d) If $N = 5 < M$, then $K(n,m) = 8$.

(e) If $6 \le N < M$, put $d = M - N$ and

$$Q(N,d) = \max\Big(\{3\} \cup \{q \ge 4 : N \ge d\lfloor q(q-2)/2 \rfloor + \kappa_d(q)\}\Big),$$

where $\kappa_d(q)$ is given in Theorem 8.2. Then

$$\boxed{K(n,m) = N + Q(N,d).}$$

Proof. Parts (a)–(d) are Theorems 3.5, 3.3 and (6.7). For $N \ge 6$, Section 6 reduces the maximum excess to the largest strict symmetric threshold, and Theorem 8.2 integerizes exactly that threshold. □

For a convenient generic inversion define

$$\kappa(d) = \begin{cases} 4, & d = 1, \\ 2, & d = 2, 3, \\ 1, & d \ge 4, \end{cases} \qquad H_d(N) = \left\lfloor \frac{N - \kappa(d)}{d} \right\rfloor,$$

$$Q_0(N,d) = \max\left\{3, \left\lceil \sqrt{2H_d(N) + 3} \right\rceil \right\}.$$

Then $Q = Q_0$ except for

$$(d, N, Q) = (3, 13, 4)$$

and, for $d = 1$,

$$(N, Q) = (6,4), (7,4), (10,5), (15,6), (20,7), (27,8), (43,10).$$

Indeed $c_q \le H$ is equivalent to $(q-1)^2 < 2H + 3$.

Corollary 9.2 (fixed-diagonal asymptotics)**.** *For every fixed $d \ge 1$,*

$$K(n, n+d) = n + \sqrt{\frac{2n}{d}} + O_d(1).$$

Proof. The exact formula has $c_q = q^2/2 + O(q)$ and a bounded correction $\kappa_d(q)$. □

10 Certificate and reproducibility

Exactly one program is used in the theorem layer:

`integer_staircase_certificate.py`.

It certifies 49 finite boundary signs with $n \ge 6$: all layers left open by the analytic tail estimates, plus seven deliberately redundant $z = 4$ positive checks in the $d = 1$ rows whose threshold is 4. In particular, both $(d, q, z) = (1, 4, 2)$ and $(1, 4, 3)$ are certified positive; no unproved monotonicity in z is used. The script also mechanically audits the finite-case ledger before evaluating any trigonometric sign. It derives a rigorous rational interval for π internally from Machin's identity, evaluates all trigonometric quantities by outward rational interval arithmetic and alternating Taylor remainders, and aborts if a required case is missing or if any interval meets zero. All geometric reductions, endpoint-defect arguments, sparse-to-symmetric domination, global realizability, low-row results, and Peano-kernel positivity are analytic and do not depend on computation.

Running

`python integer_staircase_certificate.py`

prints `PASS` together with the number of certified signs and the smallest exact interval distance from zero.

11 Concluding remarks

The exact answer separates two levels of structure. The global geometry first reduces arbitrary intersections to the strict symmetric trigonometric thresholds (5.11) and (5.15); the Peano-kernel argument then shows that their finite integer content is governed by the quadratic skeleton

$$c_q = \left\lfloor \frac{q(q-2)}{2} \right\rfloor$$

plus the bounded correction $\kappa_d(q)$ of Theorem 8.2. The exceptional boosts listed after Theorem 9.1 are therefore genuine low-layer curvature effects rather than artifacts of numerical optimization.

The larger classification question in Kagey's Problem 93 remains distinct: this paper determines the maximum possible number of sides, not all combinatorial intersection classes.

References

[1] P. Kagey, Problem 93, in *Open problem collection*, related question 1: maximum number of sides in the intersection of two unit regular polygons, available at `https://peterkagey.com/problems/093/`, accessed September 21, 2026.

[2] J. O'Rourke, *Computational Geometry in C*, 2nd ed., Cambridge University Press, Cambridge, 1998; see Section 7.6 for convex-polygon intersection.

[3] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, 2nd expanded ed., Encyclopedia of Mathematics and its Applications, vol. 151, Cambridge University Press, Cambridge, 2014. `https://doi.org/10.1017/CBO9781139003858`.